Predicting Relative Acidities

While a pKa table is more conclusive, it is possible to predict *relative* acidities simply based upon molecular structure. In order to do this, four structural characteristics must be analyzed (*in this order*):

#1: <u>Identity of the atom the proton is bonded to</u>

Within the same row, the more electronegative atom possesses the more acidic bond. This is due in part to the fact that the resulting conjugate base bears a negative charge on the most electronegative atom (which we would expect to be preferred). Here we compare the acidity of methanol (CH_3OH) to methylamine (CH_3NH_2):

$$CH_3\mathbf{O}H \quad + \quad H_2O \quad \rightleftharpoons \quad CH_3\mathbf{O}^{\ominus} \quad + \quad H_3O^{\oplus}$$

$$CH_3\mathbf{N}H_2 \quad + \quad H_2O \quad \rightleftharpoons \quad CH_3\overset{\ominus}{\mathbf{N}}H \quad + \quad H_3O^{\oplus}$$

Since **O** is *more electronegative* than **N**, the conj. base with a negative charge on **O** is *more stable* than the conj. base with a negative charge on **N**. ***Stronger acids have more stable conjugate bases***, therefore CH_3OH is a stronger acid than CH_3NH_2.

Within the same column, the size of the atom is important as the larger atoms are more polarizeable and therefore can support the resulting negative charge better than a smaller atom. Here we compare the acidity of H_2O and H_2S:

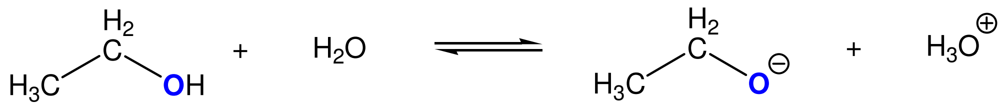
$$CH_3\mathbf{O}H \quad + \quad H_2O \quad \rightleftharpoons \quad CH_3\mathbf{O}^{\ominus} \quad + \quad H_3O^{\oplus}$$

$$CH_3\mathbf{S}H \quad + \quad H_2O \quad \rightleftharpoons \quad CH_3\mathbf{S}^{\ominus} \quad + \quad H_3O^{\oplus}$$

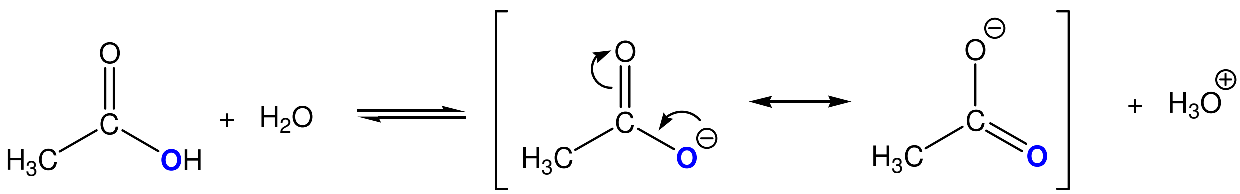
In addition to larger atoms having weaker bonds in general, the smaller carge-to-size ratio of $\mathbf{S}^{\ominus}$ increases the stability of $CH_3\mathbf{S}^{\ominus}$. Therefore, $CH_3\mathbf{S}H$ is more acidic than $CH_3\mathbf{O}H$.

#2: Identify all possible resonance structures

In general, the more resonance forms that exist for the conjugate base, the more stable it will be. In addition to this, resonance forms that put the negative charge on the more electronegative atom are more stable. As stated before, ***stronger acids have more stable conj. bases.*** This principle is illustrated by the following examples:

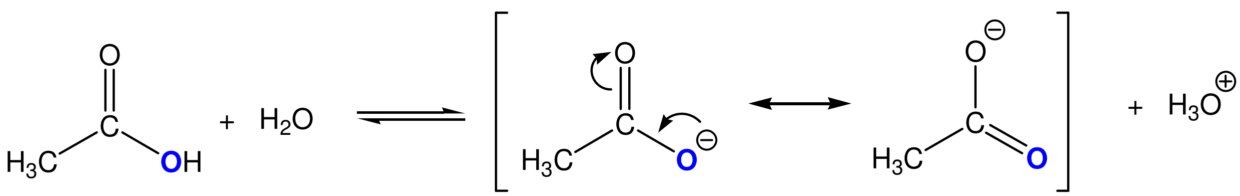


no resonance possible



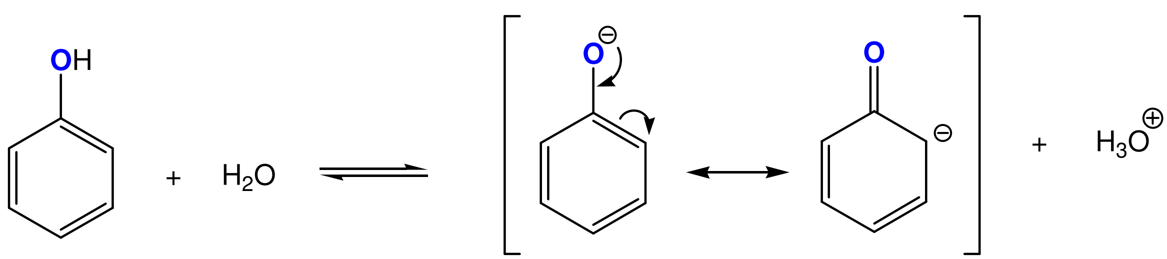
two possible resonance structures

Resonance serves to delocalize the negative charge and thereby diminishes the overall charge felt by the **O** atom. This is a stabilizing effect, therefore acetic acid (CH_3COOH; pKa = 4.74) is a stronger acid than ethanol (CH_3COH; pKa = 16).



acetic acid

*resonance puts negative charge on **O***



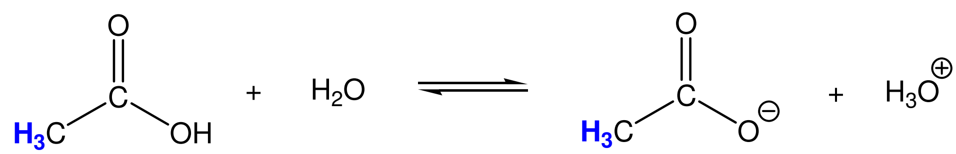
phenol

*resonance puts negative charge on **C***
*(less electronegative than **O**)*

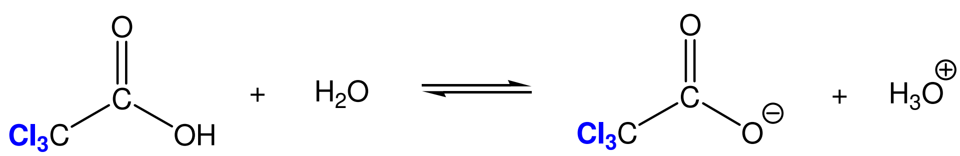
Resonance in the conj. base of acetic acid puts the negative charge on the more electronegative **O** atom. Resonance in phenol places the negative charge on a **C** atom thereby creating a less stable conj. base. We therefore predict that phenol (pKa = 9.95) is a weaker acid than acetic acid (pKa = 4.74).

#3: Inductive effects

In general, electronegative atoms or electron withdrawing groups (EWGs) that are in close proximity to the proton, result in a stronger acid.



pKa = 4.74



pKa = 0.65

In both cases the negative charge is on oxygen *and* the same resonance structures exist for both conj. bases. Since **Cl** is an electronegative atom (and therefore electron withdrawing), the charge present in the second conj. base is delocalized. This "*inductive*" effect stabilizes the conj. base. As mentioned before, ***stronger acids have more stable conj. bases.*** This effect is enhanced by the presence of more electronegative atoms (or EWGs) and decreases over distance.

#4: Hybridization of C atom

When the only bonds present in the molecule are C-H bonds, the hybridization (sp^3, sp^2, or sp) determines which C-H bond is the most acidic. The trend observed is as follows:

$$sp > sp^2 > sp^3$$

Therefore, we expect alkyne C-H bonds to be more acidic than alkene C-H bonds which are more acidic than alkane C-H bonds. This is illustrated by looking at the relative pKas for the C_2 variants of each hydrocarbon:

	$HC{\equiv}CH$	$H_2C{=}CH_2$	$H_3C\text{-}CH_3$
pKa	26	45	62